Experiment 12: Write a python program to Tic-Tac-Toe Game Using Python

Aim:

To write a Python program to implement the Tic-Tac-Toe game for two players.

Theory:

Tic-Tac-Toe is a two-player game played on a 3 × 3 grid. Player X and Player O take turns marking empty cells. The player who places three marks in a row, column, or diagonal wins the game. If all cells are filled without a winner, the game ends in a draw.

Algorithm:

1. Start the program.
2. Create a 3 × 3 game board.
3. Display the board.
4. Ask the current player to enter a row and column.
5. Check whether the selected cell is empty.
6. Check for a winning condition.
7. If the board is full, declare a draw.
8. Otherwise, switch the player and repeat.
9. Stop the game.

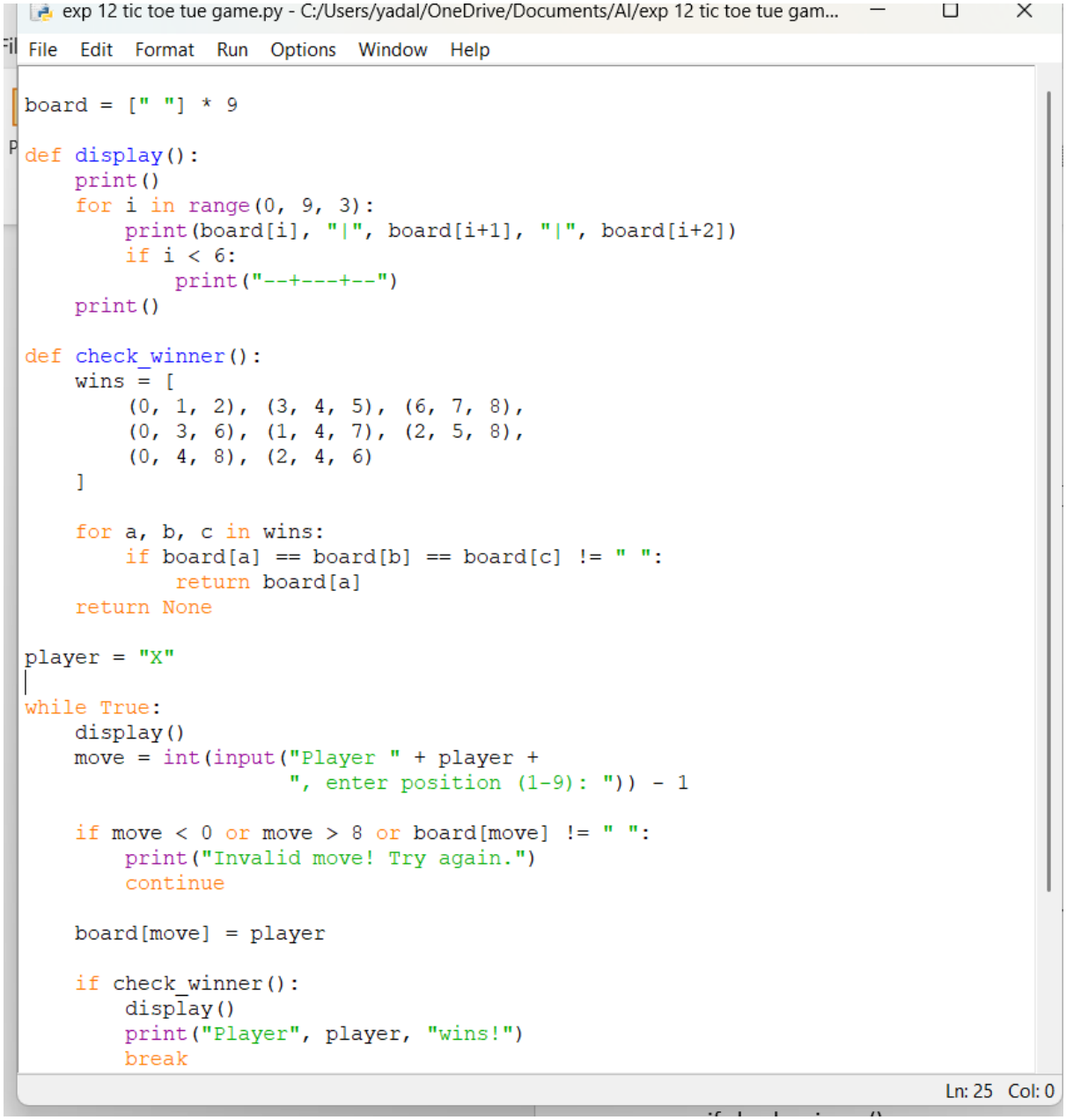
Python Program:

```
board = [" "] * 9

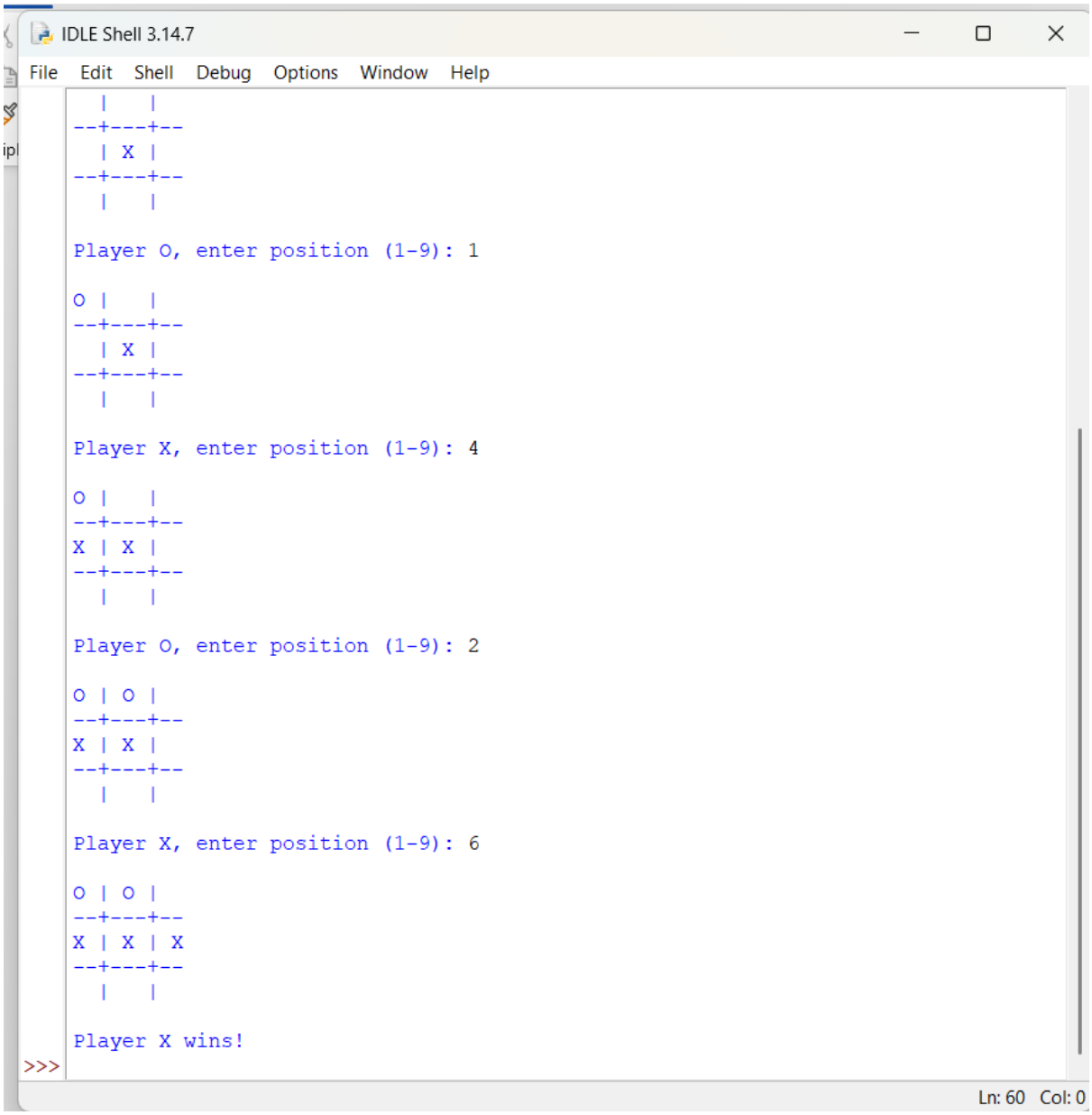
def display():
    print()
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
        if i < 6:
            print("--+---+--")
    print()

def check_winner():
    wins = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
```

```
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]
    for a, b, c in wins:
        if board[a] == board[b] == board[c] != " ":
            return board[a]
    return None
player = "X"
while True:
    display()
    move = int(input("Player " + player +
                     ", enter position (1-9): ")) - 1
    if move < 0 or move > 8 or board[move] != " ":
        print("Invalid move! Try again.")
        continue
    board[move] = player
    if check_winner():
        display()
        print("Player", player, "wins!")
        break
    if " " not in board:
        display()
        print("Match Draw!")
        break
    player = "O" if player == "X" else "X"Sample
```


exp 12 tic toe tue game.py - C:/Users/yadal/OneDrive/Documents/AI/exp 12 tic toe tue gam...
File Edit Format Run Options Window Help
board = [" "] * 9
def display():
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print(board[i], "|", board[i+1], "|", board[i+2])
if i < 6:
print("--+---+--")
print()
def check_winner():
wins = [
(0, 1, 2), (3, 4, 5), (6, 7, 8),
(0, 3, 6), (1, 4, 7), (2, 5, 8),
(0, 4, 8), (2, 4, 6)
]
for a, b, c in wins:
if board[a] == board[b] == board[c] != " ":
return board[a]
return None
player = "X"
while True:
display()
move = int(input("Player " + player +
", enter position (1-9): ")) - 1
if move < 0 or move > 8 or board[move] != " ":
print("Invalid move! Try again.")
continue
board[move] = player
if check_winner():
display()
print("Player", player, "wins!")
break
Ln: 25 Col: 0

Output:



Result:

Thus, the Tic-Tac-Toe game was successfully implemented in Python using a list, loops, conditional statements, and functions.